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Portfolio

Professional Summary

Production AI Engineer with expertise in building scalable GenAI systems, optimizing LLM inference, and deploying multi-agent architectures. Specialized in advanced RAG systems, model quantization, and LLMOps with hands-on experience in PyTorch, vLLM, and distributed training. Proven track record of reducing inference costs by 60% and improving model accuracy by 40% through production-grade optimizations.

Education

Adichunchanagiri Institute of Technology

Bachelor of Engineering in Computer Science – **CGPA: 7.8**

Aug. 2019 – Jun. 2023

Chikkamagaluru, Karnataka

- Relevant Coursework:** Advanced Algorithms, Deep Learning, NLP, Machine Learning, Database Systems, Distributed Systems, Software Engineering

Experience

Solize India Private Limited

Aug 2024 – Present

AI Project Engineer

Bangalore, Karnataka

- Architected and deployed production GraphRAG system using Neo4j and Qdrant, improving complex query accuracy by **40%** over baseline RAG through multi-hop reasoning and knowledge graph integration.
- Fine-tuned Llama-3 8B using QLoRA and DPO alignment on 10K+ domain-specific examples, achieving **35%** improvement on task-specific benchmarks while reducing training time by **50%** with Unsloth.
- Implemented production LLMOps stack with LangSmith observability, Guardrails AI for safety, and RAGAS evaluation achieving **98%+ safety rating** and automated quality monitoring across 100K+ daily queries.
- Built multi-agent orchestration system using LangGraph with 3 specialized agents (researcher, analyzer, synthesizer), reducing manual analysis time by **70%** through intelligent task decomposition and parallel execution.
- Engineered data pipelines using Polars and DuckDB processing **10M+ records** in **5 minutes**, with automated data quality checks using Great Expectations preventing **100%** of downstream errors.

Spektra Systems

Mar. 2024 – Jun. 2024

Software Development Engineer Intern (Gen AI Focus)

Bangalore, Karnataka

- Explored and integrated Gen AI APIs (OpenAI, Azure OpenAI) into internal tools to automate content generation, summarization, and code suggestions.
- Created a prototype GenAI assistant using LangChain and GPT-4 for automated internal documentation search and smart reply generation.
- Developed web applications using ASP.NET, C#, and ADO.NET, integrating Gen AI-powered features such as auto-complete and context-aware suggestions.

Fundamentals

Jan. 2023 – Mar. 2023

DevOps Intern

Chikkamagaluru, Karnataka

- Acquired experience in both software development and IT operations, actively contributing to the DevOps lifecycle, leading to a 15% increase in overall team productivity.
- Implemented continuous integration and continuous delivery (CI/CD) pipelines, automating code deployments, which reduced deployment time by 60%.
- Managed AWS cloud infrastructure, administered Linux systems, created Docker containers, deployed Kubernetes orchestration, and configured Jenkins for automated builds and deployments, improving operational efficiency by 35%.

Projects

Production Multi-Tenant RAG Platform | Python, LangGraph, Qdrant, Neo4j, FastAPI

Jan. 2026

- Built enterprise GraphRAG platform serving **1000+ concurrent users** with hybrid search (BGE-M3 + BM25 + Cohere rerank), knowledge graph reasoning, and semantic caching achieving **90%** cache hit rate.
- Implemented multi-tenant data isolation with namespace-based vector store sharding and RBAC, ensuring secure data segregation across **50+ organizations**.
- Deployed with comprehensive observability using LangSmith, Prometheus metrics, and automated RAGAS evaluation showing **95%+ accuracy** on domain benchmarks.
- Integrated with Slack and Teams for conversational AI interface, processing **100K+ queries/month** with p95 latency **300ms**.

Autonomous Multi-Agent Code Review System | LangGraph, Anthropic Claude, GitHub API, PostgreSQL Dec. 2025

- Developed multi-agent system with 3 specialized agents (reviewer, tester, optimizer) using LangGraph for stateful orchestration, finding **80%+ of issues** caught in human review.
- Implemented tool integration with linters (Ruff, Pyright), test runners (pytest), and static analyzers, automatically generating test cases with **95%** validity rate.

- Created human-in-the-loop approval workflow with memory persistence using PostgreSQL, reducing code review time by **60%** while maintaining quality standards.
- Deployed as GitHub App with webhook integration, automatically analyzing **500+ PRs/month** across multiple repositories.

Domain-Expert LLM with DPO Alignment | *PyTorch, Unsloth, TRL, Distilabel, W&B*

Nov. 2025

- Fine-tuned Llama-3 8B on 10K synthetic examples using QLoRA (4-bit) and DPO alignment, achieving **30%** improvement on domain benchmarks vs base model.
- Built synthetic data generation pipeline using Distilabel and GPT-4 for evol-instruct patterns, creating high-quality training data at **1/10th** the cost of human annotation.
- Deployed production inference using vLLM with speculative decoding achieving **2x speedup** and serving via FastAPI with automatic batching handling **100+ concurrent requests**.
- Implemented comprehensive evaluation suite with RAGAS, DeepEval showing **98%** safety rating and **5%** hallucination rate on adversarial tests.

High-Performance LLM Inference Platform | *vLLM, TensorRT-LLM, Kubernetes, Prometheus*

Oct. 2025

- Built production inference system serving Llama-3 70B (4-bit GPTQ) with vLLM achieving **10-20x throughput** vs naive serving through PagedAttention and continuous batching.
- Implemented full observability stack with LangSmith tracing, Prometheus metrics, and Guardrails AI for input/output validation, ensuring **99.9%** uptime and **100%** PII redaction.
- Deployed on Kubernetes with auto-scaling based on queue depth, handling peak loads of **10K+ requests/hour** while maintaining p95 latency **2 seconds**.
- Achieved **60%** cost reduction vs OpenAI API through model quantization, batching optimization, and spot instance utilization on cloud GPUs.

Technical Skills

Languages & Core: Python (Advanced, FastAPI), Java, JavaScript

Deep Learning: PyTorch 2.x, TensorFlow/Keras, Hugging Face Transformers

LLM & GenAI: LangChain, LangGraph, LlamaIndex, vLLM, LoRA/QLoRA, DPO/ORPO, Unsloth

Vector & Graph DBs: Qdrant, Pinecone, Neo4j, ChromaDB, pgvector

Data Engineering: Polars, DuckDB, Pandas, NumPy, Apache Arrow, Delta Lake, Parquet

MLOps & Monitoring: Weights & Biases, MLflow, LangSmith, Arize Phoenix, RAGAS, DeepEval

Cloud & Infrastructure: AWS (S3, EC2, RDS, SQS), GCP, Docker, Kubernetes, Serverless

Search & Retrieval: Hybrid Search, BGE-M3, E5, Cohere Rerank, BM25, Semantic Caching, Redis

DevOps & Tools: Git, GitHub Actions, CI/CD, Grafana,